Tutorial - 08

Applicant ID	x_1 Credit Score	x_2 Annual Income	x_3 Employees	x_4 loan amount	Approved
1	750	80	8	200	A
2	680	35	2	150	R
3	780	95	10	250	A
4	580	30	1	180	R
5	720	75	7	180	A
6	620	40	3	160	R

	y_1	y_2	y_3	y_4	
T_1	700	70	6	190	A
T_2	610	38	2	155	R
	y_1	y_2	y_3	y_4	
T_3	770	90	9	230	A

$$D_{T} = \sqrt{(x_1-y_1)^2 + (x_2-y_2)^2 + (x_3-y_3)^2 + (x_4-y_4)^2}$$

$$= \sqrt{(750-700)^2 + (80-70)^2 + (8-2)^2 + (200-190)^2}$$

$$= 52$$

$$2 = \sqrt{(680-700)^2 + (35-70)^2 + (2-6)^2 + (150-100)^2}$$

$$= 113.3$$

$$3 = \sqrt{(780-700)^2 + (95-70)^2 + (10-6)^2 + (250-190)^2}$$

$$= 103.15$$

$$4 = \sqrt{(580-700)^2 + (30-70)^2 + (1-6)^2 + (180-190)^2}$$

$$= 126.98$$

$5 = \sqrt{(720-700)^2 + (75-70)^2 + (7-6)^2 + (180-190)^2}$

$= 22.9$

$6 = \sqrt{(620-700)^2 + (40-70)^2 + (3-6)^2 + (160-190)^2}$

$= 90.60$

Next step do with T_2 y values are T_2

$1 = \sqrt{(650-610)^2 + (80-38)^2 + (8-2)^2 + (200-155)^2}$

$= 153.7$

$2 = \sqrt{(680-610)^2 + (35-38)^2 + (2-2)^2 + (150-155)^2}$

$= 11.57$

$3 = \sqrt{(780-610)^2 + (95-38)^2 + (10-2)^2 + (250-155)^2}$

$= 203$

$4 = \sqrt{(580-610)^2 + (30-38)^2 + (1-2)^2 + (180-155)^2}$

$= 39.87$

$5 = \sqrt{(720-610)^2 + (75-38)^2 + (7-2)^2 + (180-155)^2}$

$= 118$

$6 = \sqrt{(620-610)^2 + (40-38)^2 + (2-3)^2 + (155-160)^2}$

$= 11.40$

Next step do with T_3 y values are T_3.

$1 = \sqrt{(750-770)^2 + (80-90)^2 + (8-9)^2 + (200-230)^2}$

$= 37.42$

$2 = \sqrt{(680-770)^2 + (35-90)^2 + (2-9)^2 + (150-230)^2}$

$= 195.89$

$3 = \sqrt{(780-770)^2 + (95-90)^2 + (10-9)^2 + (250-230)^2}$

$$4 = \sqrt{(380-770)^2 + (30-90)^2 + (1-9)^2 + (180-230)^2}$$

$$= 205.58$$

$$5 = \sqrt{(420-770)^2 + (75-90)^2 + (7-9)^2 + (180-230)^2}$$

$$= 72.31$$

$$6 = \sqrt{(620-770)^2 + (40-90)^2 + (3-9)^2 + (160-230)^2}$$

$$= 173.02$$